Software Engineering

Lab-08

22-10-2026

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Static Code Analysis

Executing the file before any changes

```
(venv) pre@fedora ~/docs/acads/sem-5/SE/lab/lab-5  % python3 restaurant_order_system.py
Is T1 pending: True
Discounted Bill: 40.5
Ticket Hash: ce499dea30cfce118f4fe85da0227e83
POS PIN: 2801
Ticket sent to kitchen printer: kitchen_orders.txt
Pending tables: 3
POS Secret active: POS_TERMINAL_SECRET_TOKEN_2026
(venv) pre@fedora ~/docs/acads/sem-5/SE/lab/lab-5  %
```

creating .txt reports before changes

```
(venv) pre@fedora ~/docs/acads/sem-5/SE/lab/lab-5  % pylint restaurant_order_system.py  > reports/before/pylint_report.txt
(venv) pre@fedora ~/docs/acads/sem-5/SE/lab/lab-5  % bandit -r restaurant_order_system.py  > reports/before/bandit_report.txt
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.14.7
(venv) pre@fedora ~/docs/acads/sem-5/SE/lab/lab-5  % flake8 restaurant_order_system.py  > reports/before/flake8_report.txt
(venv) pre@fedora ~/docs/acads/sem-5/SE/lab/lab-5  %
```

viewing all the reports

```
(venv) pre@fedora ~/docs/acads/sem-5/SE/lab/lab-5  % cat reports/before/pylint_report.txt  | tail -20
cat reports/before/bandit_report.txt  | tail -40
cat reports/before/flake8_report.txt
restaurant_order_system.py:34:4:  R1703: The if statement can be replaced with 'return bool(test)' (simplifiable-if-statement)
restaurant_order_system.py:34:4:  R1705: Unnecessary "else" after "return", remove the "else" and de-indent the code inside it (no-else-return)
restaurant_order_system.py:31:4:  W0612: Unused variable 'service_tax' (unused-variable)
restaurant_order_system.py:40:0:  C0116: Missing function or method docstring (missing-function-docstring)
restaurant_order_system.py:40:0:  C0103: Function name "calculateDiscountedBill"  doesn't conform to snake_case naming style (invalid-name)
restaurant_order_system.py:42:16:  W0123: Use of eval (eval-used)
restaurant_order_system.py:41:4:  W0612: Unused variable 'base_bill' (unused-variable)
restaurant_order_system.py:46:0:  C0116: Missing function or method docstring (missing-function-docstring)
restaurant_order_system.py:50:0:  C0116: Missing function or method docstring (missing-function-docstring)
restaurant_order_system.py:54:0:  C0116: Missing function or method docstring (missing-function-docstring)
restaurant_order_system.py:55:8:  W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
restaurant_order_system.py:55:8:  R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
restaurant_order_system.py:61:0:  C0116: Missing function or method docstring (missing-function-docstring)
restaurant_order_system.py:61:24:  W0621: Redefining name 'orders' from outer scope (line 8) (redefined-outer-name)
restaurant_order_system.py:63:8:  W0612: Unused variable 't' (unused-variable)
restaurant_order_system.py:6:0:  W0611: Unused import math (unused-import)

-------------------------------------------------------------------
Your code has been rated at 4.81/10 (previous run: 4.81/10, +0.00)


--------------------------------------------------
>> Issue: [B311:blacklist] Standard pseudo-random generators are not suitable for security/cryptographic purposes.
   Severity: Low   Confidence: High
   CWE: CWE-330 (https://cwe.mitre.org/data/definitions/330.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b311-random
   Location: ./restaurant_order_system.py:51:11
50      def generate_pos_pin():
51          return random.randint(1000, 9999)
52
```

```
>> Issue: [B605:start_process_with_a_shell]  Starting a process with a shell, possible injection detected, security issue.
   Severity: High   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b605_start_process_with_a_shell.html
   Location: ./restaurant_order_system.py:58:4
57          f.close()
58          os.system("echo Ticket sent to kitchen printer: " + filename)
59

--------------------------------------------------

Code scanned:
        Total lines of code: 61
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 2
                Medium: 1
                High: 2
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 5
Files skipped (0):
```

static analysis using bandit

```
(venv) pre@fedora ~/docs/acads/sem-5/SE/lab/lab-5/L5_49_restaurant_order_system   % bandit -r restaurant_order_system.py  | tee reports/after/bandit_repo
rt.txt
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.14.7
Run started:2026-10-22  09:00:10.584288+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 59
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
```

static analysis using flake

```
(venv) pre@fedora ~/docs/acads/sem-5/SE/lab/lab-5/L5_49_restaurant_order_system   % bandit -r restaurant_order_system.py  | tee reports/after/bandit_repo
rt.txt
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.14.7
Run started:2026-10-22  09:02:48.573921+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 68
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
```